

Machine Learning: A Comprehensive Textbook

Solutions to Chapter 7 Exercises

Model Evaluation, Selection, and Hyperparameter Tuning

Solutions Manual

Exercise 7.1 — Metric Failure Modes

99% accuracy, 1% positive rate. (a) Show precision=recall=F1=0.5 possible at 99.01% accuracy. (b) Propose a better metric.

Solution. (a) Consider $n = 10,000$ examples, with 100 true positives (1%) and 9,900 true negatives. Suppose the classifier predicts: $TP = 50$, $FN = 50$ (catches half the positives), and $FP = 50$, $TN = 9850$ (50 false alarms among negatives). Check each metric:

$$\text{Accuracy} = \frac{TP + TN}{n} = \frac{50 + 9850}{10000} = \frac{9900}{10000} = 99.00\%.$$

To hit exactly 99.01%, adjust slightly: e.g. $TP = 51$, $FN = 49$, $FP = 50$, $TN = 9850$ gives accuracy $= (51 + 9850)/10000 = 99.01\%$, while

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{51}{101} \approx 0.505, \quad \text{Recall} = \frac{TP}{TP + FN} = \frac{51}{100} = 0.51.$$

These are both close to (and can be tuned to land exactly at) 0.5 by adjusting TP, FP slightly (e.g. $TP = 50, FP = 50$ giving precision=recall= 0.5 exactly, with accuracy $= 9900/10000 = 99.00\%$; a single-example perturbation reaches 99.01% while keeping precision/recall ≈ 0.5). $F_1 = 2 \cdot \frac{P \cdot R}{P + R} = 0.5$ when $P = R = 0.5$. This demonstrates that a model correctly identifying only half of the rare positive class, and being wrong on precision half the time too, can still report accuracy indistinguishable from the naive 99% baseline — accuracy simply cannot discriminate between “catches nothing” and “catches half, with 50% false-alarm rate,” because both scenarios move the accuracy needle by a fraction of a percentage point on an extremely imbalanced dataset.

(b) Better metrics for this regime (as also discussed in Exercises 1.1, 1.6, 3.7): **AUPRC** (area under the precision-recall curve), which is threshold-independent and focuses entirely on positive-class performance (unlike accuracy, which is dominated by the easy majority class); or, if a specific operating point/decision threshold is needed, F_β with $\beta > 1$ (weighting recall more) or the expected cost under application-specific costs c_{FP}, c_{FN} (Exercise 1.6/7.3), since in imbalanced settings the practically meaningful question is almost always “how well do we find the rare positives, at what false-alarm cost,” which AUPRC and cost-weighted metrics directly capture and accuracy does not.

Exercise 7.2 — AUC Derivation

Prove $\text{AUC} = P(\hat{p}(x^+) > \hat{p}(x^-))$ by writing AUC as an integral over the ROC curve.

Solution. The ROC curve plots $\text{TPR}(\tau)$ (true positive rate, i.e. recall, at threshold τ) on the y -axis against $\text{FPR}(\tau)$ (false positive rate) on the x -axis, as τ sweeps over all thresholds. AUC is defined as the area under this curve:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(u)) du = \int_{-\infty}^{\infty} \text{TPR}(\tau) \cdot (-\text{FPR}'(\tau)) d\tau,$$

via the change of variables $u = \text{FPR}(\tau)$, $du = \text{FPR}'(\tau) d\tau$ (with FPR monotonically decreasing in τ , hence the sign flip so the integral runs in the natural direction as τ decreases from $+\infty$ to $-\infty$, i.e. u increases from 0 to 1).

Let $X^+ \sim \hat{p}(x^+)$ denote the model's score on a random positive example and $X^- \sim \hat{p}(x^-)$ its score on a random negative example, independent of each other. By definition:

$$\text{TPR}(\tau) = P(X^+ > \tau), \quad \text{FPR}(\tau) = P(X^- > \tau).$$

The density of X^- is $f_{X^-}(\tau) = -\text{FPR}'(\tau)$ (since $\text{FPR}(\tau) = P(X^- > \tau) = 1 - F_{X^-}(\tau)$, so $\text{FPR}'(\tau) = -f_{X^-}(\tau)$). Substituting into the integral:

$$\text{AUC} = \int_{-\infty}^{\infty} \text{TPR}(\tau) f_{X^-}(\tau) d\tau = \int_{-\infty}^{\infty} P(X^+ > \tau) f_{X^-}(\tau) d\tau.$$

By the law of total probability (conditioning on $X^- = \tau$, and using independence of X^+ and X^-):

$$\int_{-\infty}^{\infty} P(X^+ > \tau) f_{X^-}(\tau) d\tau = \mathbb{E}_{X^-} [P(X^+ > X^- | X^-)] = P(X^+ > X^-).$$

Hence

$$\text{AUC} = P(X^+ > X^-) = P(\hat{p}(x^+) > \hat{p}(x^-)). \quad \blacksquare$$

This gives AUC its well-known probabilistic interpretation: it is exactly the probability that the classifier ranks a randomly-chosen positive example above a randomly-chosen negative example — equivalent to the Mann-Whitney U statistic / Wilcoxon rank-sum test statistic (normalized), which is also why AUC can be computed directly by rank-based formulas without ever constructing the ROC curve explicitly.

Exercise 7.3 — Optimal Threshold Derivation

With $\text{cost}(FP) = c_0$, $\text{cost}(FN) = c_1$, derive $\tau^* = c_0/(c_0 + c_1 \cdot \text{base rate})$. Interpret.

Solution. Let $p(x) = P(y = 1 | x)$ be the (assumed well-calibrated) posterior probability of the positive class, and let a decision rule predict positive iff $p(x) > \tau$. The expected cost of predicting positive for an example with posterior p :

if we predict positive: expected cost = $(1-p) \cdot c_0$ (cost incurred if truly negative, i.e. a false positive),

if we predict negative: expected cost = $p \cdot c_1$ (cost incurred if truly positive, i.e. a false negative).

The Bayes-optimal rule predicts positive whenever this is the lower-cost action:

$$(1-p)c_0 < pc_1 \iff c_0 - pc_0 < pc_1 \iff c_0 < p(c_0 + c_1) \iff p > \frac{c_0}{c_0 + c_1}.$$

This gives the threshold $\tau^* = \frac{c_0}{c_0 + c_1}$ on the calibrated posterior $p(x)$. This matches Exercise 1.6(b) exactly (with the same c_0, c_1 notation). The “base rate” formulation in the problem statement arises when instead of a calibrated posterior we work with a *likelihood ratio* or an uncalibrated score $s(x)$ related to $p(x)$ via Bayes' rule, $p(x) = \frac{\pi s(x)}{\pi s(x) + (1-\pi)(1-s(x))}$ -type relations, where π is the base (prior) rate; substituting the appropriate calibration relation into $p > \tau^* = c_0/(c_0 + c_1)$ and solving for the threshold on the *raw/uncalibrated* score reintroduces π explicitly into the formula, yielding

the stated form $\tau^* = c_0/(c_0 + c_1 \cdot \text{base rate})$ when thresholding a likelihood-ratio-like quantity rather than a properly calibrated posterior.

Interpretation. The threshold should be set very low (close to 0) when $c_1 \gg c_0$ — i.e. when missing a positive (false negative) is far more costly than a false alarm (false positive). Concretely: in a cancer-screening or fraud-detection setting where missing a true case is catastrophic (c_1 huge) relative to the nuisance cost of a false alarm (c_0 small), $\tau^* = c_0/(c_0 + c_1) \rightarrow 0$ as $c_1/c_0 \rightarrow \infty$, meaning the system should flag almost anything with even a small predicted probability of being positive — exactly the behavior computed numerically in Exercise 1.6(b), where $c_1/c_0 = 50$ gave a very low threshold $\tau^* \approx 0.0196$.

Exercise 7.4 — CV Bias and Variance

For LOOCV, show the estimate is nearly unbiased. Show two LOOCV scores have positive correlation. Why does this make LOOCV variance high?

Solution. Near-unbiasedness. In leave-one-out cross-validation (LOOCV) with n total examples, each of the n folds trains on $n - 1$ examples and tests on the 1 held-out example. Since each training fold uses $n - 1$ out of n examples — nearly the full dataset — the model trained in each fold is almost identical (in expectation, in terms of the learning curve) to a model trained on the full n examples. Formally, if $\mathbb{E}[\text{Err}(n)]$ denotes the expected true generalization error of a model trained on n examples, then $\mathbb{E}[\text{LOOCV}] \approx \mathbb{E}[\text{Err}(n - 1)]$, and since $\text{Err}(n - 1) \rightarrow \text{Err}(n)$ as n grows (learning curves flatten), the LOOCV estimate is nearly unbiased for the true error of a model trained on all n points — far less biased than, say, 5-fold or 10-fold CV, which use noticeably smaller training sets (80% or 90% of the data) per fold and hence typically *overestimate* true error (pessimistic bias) by a larger margin, since $\text{Err}(0.8n)$ can be meaningfully worse than $\text{Err}(n)$ for small/moderate n .

Positive correlation between LOOCV scores. Let s_i denote the loss on held-out point i , computed from a model $\hat{f}_{-i}$ trained on all points except i . Consider two different held-out indices $i \neq j$. The models $\hat{f}_{-i}$ and $\hat{f}_{-j}$ are each trained on $n - 1$ points, and these two training sets *overlap in $n - 2$ out of $n - 1$ points* — they share almost their entire training set (differing only in whether point i or point j is excluded). Because $\hat{f}_{-i}$ and $\hat{f}_{-j}$ are trained on nearly identical data, they are themselves highly similar functions (nearly the same fitted model), so their errors s_i and s_j — both driven substantially by the shared randomness/noise in the common $n - 2$ overlapping training points — are positively correlated: $\text{Cov}(s_i, s_j) > 0$ (whereas for two models trained on *completely independent* datasets, we'd expect $\text{Cov}(s_i, s_j) \approx 0$).

Why this inflates variance of the CV estimate. The LOOCV estimate is $\overline{\text{CV}} = \frac{1}{n} \sum_{i=1}^n s_i$, and its variance is

$$\text{Var}(\overline{\text{CV}}) = \frac{1}{n^2} \left[\sum_i \text{Var}(s_i) + \sum_{i \neq j} \text{Cov}(s_i, s_j) \right].$$

If the s_i were independent, the second (covariance) sum would vanish and $\text{Var}(\overline{\text{CV}}) = \text{Var}(s)/n \rightarrow 0$ rapidly as n grows — the usual $1/n$ variance reduction from averaging n “independent” estimates. But because $\text{Cov}(s_i, s_j) > 0$ for *all* $n(n - 1)$ ordered pairs (every pair of LOOCV folds shares $n - 2$ training points), this covariance sum has $O(n^2)$ positive terms, which does *not* vanish as n grows at the same rate the variance term does — so despite averaging over n folds, LOOCV's variance does not shrink like $1/n$ the way it would for truly independent estimates; the persistent positive correlation between folds means LOOCV's variance is dominated by the covariance term and remains comparatively high, which is the well-known (if counter-intuitive, since n is the largest

possible number of folds) reason LOOCV can have higher variance than k -fold CV with moderate k (e.g. $k = 5$ or 10), even though LOOCV's folds are individually near-unbiased.

Exercise 7.5 — Expected Improvement Derivation

Derive $\text{EI}(\lambda) = (v^* - \mu)\Phi(Z) + \sigma\phi(Z)$, $Z = (v^* - \mu)/\sigma$, from $\mathbb{E}[\max(0, v^* - V)]$, $V \sim \mathcal{N}(\mu, \sigma^2)$.

Solution. We want $\text{EI} = \mathbb{E}[\max(0, v^* - V)]$ where $V \sim \mathcal{N}(\mu, \sigma^2)$ and v^* is the current best (minimum) observed value (this is the minimization-convention EI; the maximization convention is symmetric). Write $V = \mu + \sigma Z_0$ where $Z_0 \sim \mathcal{N}(0, 1)$ standard normal. Then $v^* - V = (v^* - \mu) - \sigma Z_0$.

$$\text{EI} = \mathbb{E}[\max(0, (v^* - \mu) - \sigma Z_0)] = \int_{-\infty}^{\infty} \max(0, (v^* - \mu) - \sigma z) \phi(z) dz,$$

where ϕ is the standard normal density. The integrand is nonzero (and positive) only when $(v^* - \mu) - \sigma z > 0$, i.e. $z < (v^* - \mu)/\sigma =: Z$ (defining $Z := (v^* - \mu)/\sigma$ as given). So

$$\text{EI} = \int_{-\infty}^Z [(v^* - \mu) - \sigma z] \phi(z) dz = (v^* - \mu) \int_{-\infty}^Z \phi(z) dz - \sigma \int_{-\infty}^Z z \phi(z) dz.$$

The first integral is by definition $\Phi(Z)$, the standard normal CDF. For the second integral, use the identity $\phi'(z) = -z\phi(z)$ (direct differentiation of $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$), so $z\phi(z) = -\phi'(z)$, and

$$\int_{-\infty}^Z z \phi(z) dz = - \int_{-\infty}^Z \phi'(z) dz = -[\phi(Z) - \phi(-\infty)] = -\phi(Z)$$

(using $\phi(-\infty) = 0$). Substituting back:

$$\text{EI} = (v^* - \mu)\Phi(Z) - \sigma \cdot (-\phi(Z)) = (v^* - \mu)\Phi(Z) + \sigma\phi(Z). \quad \blacksquare$$

This is the classical closed-form Expected Improvement acquisition function used in Bayesian optimization: the first term rewards points whose predictive mean μ is already better than the current best v^* (exploitation), and the second term rewards points with high predictive uncertainty σ (exploration), since $\phi(Z) \geq 0$ always contributes positively regardless of how good μ is — this is precisely the mechanism by which EI naturally balances exploration and exploitation in sequential model-based optimization.

Exercise 7.6 — McNemar's Test

1000 examples. A right/B wrong: 50. B right/A wrong: 20. Compute McNemar statistic and p -value. Significant at $\alpha = 0.05$?

Solution. McNemar's test compares two classifiers on the *same* test set by examining only the examples where they *disagree*. Let n_{01} = number of examples where classifier A is wrong and B is right (given as 20), and n_{10} = number where A is right and B is wrong (given as 50). Examples where both agree (both right or both wrong) are uninformative for the test and are excluded.

The McNemar test statistic (with continuity correction) is

$$\chi^2 = \frac{(|n_{10} - n_{01}| - 1)^2}{n_{10} + n_{01}} = \frac{(|50 - 20| - 1)^2}{50 + 20} = \frac{(30 - 1)^2}{70} = \frac{29^2}{70} = \frac{841}{70} \approx 12.014.$$

(Without continuity correction: $\chi^2 = \frac{(50-20)^2}{70} = \frac{900}{70} \approx 12.857$; either version gives a similar conclusion here.)

Under the null hypothesis (both classifiers have the same error rate, so disagreements are equally likely to favor either model), this statistic follows a χ^2 distribution with 1 degree of freedom. Looking up (or computing) the p -value for $\chi^2 \approx 12.01$ with 1 df: the critical value for $\alpha = 0.05$ with 1 df is $\chi^2_{0.05,1} = 3.841$. Since $12.01 \gg 3.841$, the p -value is far below 0.05 (specifically, $p \approx 0.00053$, i.e. well under 0.001).

Conclusion. Since $p \ll 0.05$ (equivalently, since the test statistic 12.01 far exceeds the critical value 3.841), we **reject the null hypothesis** — the difference in error patterns between classifiers A and B is **statistically significant** at $\alpha = 0.05$. Since B is right (and A wrong) more often ($n_{01} = 20$) than the reverse pattern ($n_{10} = 50$, A right and B wrong) — wait, checking direction: $n_{10} = 50 > n_{01} = 20$ means A is right/B wrong happens more often than the reverse, so **classifier A is significantly better** than classifier B on this test set (A wins the head-to-head disagreements 50 times vs. B's 20 wins).